uct features. While requirements are a little vague, product features give us specific figures that form the basis of product design. The filled QFD chart for the charge converter (Fig. 3) shows this process.

For the sake of commercial confidentially I have changed the competitors' product names and instead called them Prod. A, Prod. B, etc. The highlighted cell indicates where we want our product to be in terms of the requirements. The triangular portion shows the corelations between different product parameters. Just below the parameters, we write the target values for the product.

We shall take a specific example to show how the product specification is determined. In the requirement, we noted that the charge converter should be able to handle the power requirement for a small family. Looking at the other benchmark products we find the rated power varies from 500W to 1,200W.

We want our product to be of premium quality and so decided that it should be able to handle 1,500W power from the alternator. It is also decided that the operating voltage of the converter will be 48V. This gives the current rating as 1500W/48V = 31.25 amps. We add a 10% safety margin and set the product specification as 35 amps and 48 volts.

The QFD process uses four such charts to describe the complete product development plan (Fig. 2). In the QFD jargon, we call these charts a house.

The first house is used to arrive at the product specifications from the product requirements. While product requirements are generic and qualitative, product specifications are more quantitative values. In the second house, we translate the product specifications into product design parameters. The third house gives us the manufacturing plan and the fourth house gives us the quality assurance plan. These four charts together describe the high-level product development plan.

Fig. 3 shows the QFD chart for the first house. Here we derive product specifications from the list of requirements. Requirements are the list of things that we want our product to perform. We list different requirements in the first column of the QFD and then take an evaluation on the priority of these requirements.

Rating 7 to 9 are reserved for requirements that are in the "must have" category. Rating 1 to 3 is a "nice to have" kind of requirement. These ratings indicate that customers would not care much about the requirement. The rest of the requirements are rated between 4 and 6.

Next, try to figure out the product features which will address the